Ice Storms

Unique meteorological phenomena that produce spectacular beauty and destruction in equal measure

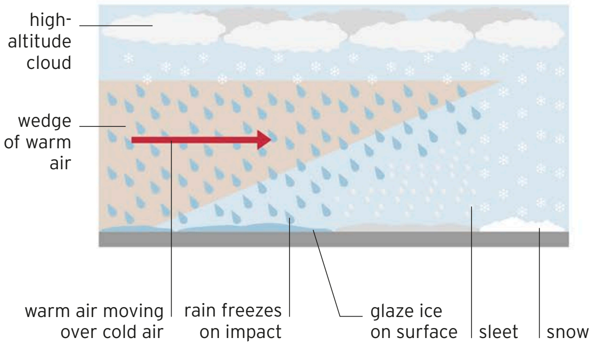
Unlike in other storms, conditions are often calm and quiet during an ice storm. But in terms of loss of life, disruption to transportation and power networks, and general mayhem, a major ice storm can produce as much chaos as a tornado or hurricane. The vital elements that cause this phenomenon are a combination of heavy rain falling from a wedge of warm air, an underlying area of supercold air, and freezing ground temperatures. When the rain hits the ground, or any subfreezing structure close to it, it freezes and forms a layer of ice known as glaze ice. As the rain persists, this glaze ice grows thicker.

Collapse under pressure

Glaze ice can be over 2 in (5 cm) thick. In 1961, a layer 8 in (20 cm) thick was measured in parts of Idaho. Since ice is 10 times as heavy as the equivalent thickness of wet snow, the effects are often dramatic. Apart from turning everything white, power lines, trees, and unstable buildings are brought down. Roads become impassable, and aircraft are grounded. In January 1998, an ice storm hit a vast area of New England and southeast Canada—a region particularly susceptible to these events. At least 44 people died, with hundreds more injured. Some 80,000 miles (129,000 km) of power cables collapsed, leaving four million people without power.

SLEET, SNOW, AND FREEZING RAIN

If snow passes first through a wedge of warm air then a shallow layer of cold air, it turns into rain, then freezes as it hits the ground. If it passes through a thicker layer of cold air beneath the warm, it partially refreezes as sleet. If it does not pass through the warm air, it lands as snow.



◁ **FROZEN FRUIT**
Thousands of apple trees in New York, which were damaged in the January 1998 ice storm, took years to recover.

Ice can **increase** the **weight** of **branches** by **30 times**